DAYANANDA SAGAR COLLEGE OF ENGINEERING
(An Autonomous Institute affiliated to Visvesvaraya Technological University (VTU), Belagavi, Approved by AICTE and UGC, Accredited by NAAC with 'A' grade & ISO 9001–2015 Certified Institution)
Shavige Malleshwara Hills, Kumaraswamy Layout, Bengaluru-560 111, India

DEPARTMENT OF ELECTRONICS & COMMUNICATION ENGINEERING

ALTERNATE ASSESSMENT TOOL (AAT-2)

Design and Simulation of a Low-Power 6T SRAM Cell Using Power-Gating Technique in Cadence Virtuoso

Submitted for the subject

Low Power VLSI Design (22EC743)

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CERTIFICATE

This is to certify that the Alternate Assessment Tool (AAT-2) entitled **"Design and Simulation of a Low-Power 6T SRAM Cell Using Power-Gating Technique in Cadence Virtuoso"** as part of **Low Power VLSI Design (22EC743)** is a bonafide work carried out by **Abhishek Shetty S S (1DS23EC006)** and **Akash L A (1DS23EC016)** in partial fulfilment of the requirements for the 7th semester of the Bachelor of Engineering in Electronics and Communication Engineering of Visvesvaraya Technological University, Belagavi, during the academic year 2026-27. The AAT report has been approved as it satisfies the academic requirements prescribed for the Bachelor of Engineering degree.

Signature of Faculty	**Signature of HOD**
Dr. Jamuna S	**Dr. Shobha K R**
Professor	Head of the Department

DECLARATION

We declare that we abide by the ethical principles and commit to professional ethics, responsibilities and norms of engineering practice. The work submitted in this report for Low Power VLSI Design (22EC743), VII Semester B.E., ECE, has been compiled by referring to relevant online and offline resources to the best of our understanding, in partial fulfilment of the requirements for the award of the degree of Bachelor of Engineering in Electronics and Communication Engineering at Dayananda Sagar College of Engineering, an autonomous institution affiliated to VTU, Belagavi, during the academic year 2026-27.

We further declare that this work has not been submitted in part or in full for any other academic purpose.

ABHISHEK SHETTY S S (1DS23EC006)
AKASH L A (1DS23EC016)

Place: Bengaluru
Date: 30 October 2026

TABLE OF CONTENTS

Chapter 1: Introduction 1
1.1 Abstract 1
1.2 Objectives 2
1.3 Introduction to SRAM 2
1.4 Organisation of the Report 3
Chapter 2: Algorithm and Workflow 4
2.1 System Architecture 4
2.2 Working Principle of 6T SRAM 5
2.3 Working Principle of Power Gating 5
2.4 Input and Control Conditions 6
2.5 Design Workflow 6
Chapter 3: Theoretical Background and Analysis 8
3.1 Introduction 8
3.2 Structure of a Conventional 6T SRAM Cell 8
3.3 SRAM Cell Operation 9
3.4 Power Consumption in SRAM 9
3.5 Power-Gating Technique 10
3.6 Working of the Proposed Power-Gated SRAM 10
3.7 Theoretical Analysis of Power Reduction 11
3.8 Virtual Supply Voltage 11
3.9 Design Considerations 12
Chapter 4: Cadence Virtuoso Implementation 13
4.1 Introduction 13
4.2 Implementation of the Conventional 6T SRAM Cell 13
4.3 Integration of Power-Gating Transistors 14
4.4 Test Bench Implementation 14
4.5 Transient Analysis Setup 15
4.6 Average Power Measurement Using the Cadence Calculator 16
Chapter 5: Simulation Results and Discussion 17
5.1 Introduction 17
5.2 Transient Response Analysis 17
5.3 Active-Mode Power Measurement 18
5.4 Sleep-Mode Power Measurement 19
5.5 Comparison of Simulation Results 19
5.6 Overall Results Discussion 20
5.7 Result Summary 20
Chapter 6: Advantages and Limitations 22
6.1 Advantages 22
6.2 Disadvantages 22
6.3 Limitations 23
Chapter 7: Conclusion and Future Scope 24
7.1 Conclusion 24
7.2 Future Scope 24
Chapter 8: References 25

LIST OF FIGURES

Fig. 2.1: Basic architecture of a conventional 6T SRAM cell. 4

Fig. 2.2: Power-gating architecture using header and footer sleep transistors. 6

Fig. 2.3: Design and simulation workflow. 7

Fig. 4.1: Conventional 6T SRAM cell implementation in Cadence Virtuoso. 13

Fig. 4.2: Power-gated 6T SRAM schematic with header and footer sleep transistors. 14

Fig. 4.3: Test bench configuration of the power-gated SRAM in Cadence Virtuoso. 15

Fig. 5.1: Transient response of Qb, instantaneous power, Q, sleep signal and virtual supply voltage. 17

Fig. 5.2: Transient response of the 6T SRAM cell showing Q, QB, WL, BL and BLB. 18

Fig. 5.3: Power measurement in Cadence Virtuoso: active-mode simulation run. 19

Fig. 5.4: Power measurement in Cadence Virtuoso: sleep-mode simulation run. 19

LIST OF TABLES

Table 2.1: Operating conditions in active and sleep modes.....6

Table 3.1: Transistors of the 6T SRAM cell and their functions.....8

Table 3.2: Comparison of active and sleep modes. 10

Table 4.1: Signals used in the simulation. 15

Table 4.2: Simulation settings.....16

Table 5.1: Comparison of active-mode and sleep-mode power. 20

CHAPTER 1
INTRODUCTION

1.1 Abstract

Static Random Access Memory (SRAM) is a key memory component in modern digital systems, processors, microcontrollers and embedded applications. SRAM offers high-speed data storage and, unlike Dynamic Random Access Memory (DRAM), does not require periodic refreshing. A conventional 6T SRAM cell consists of six MOS transistors: two cross-coupled inverters and two access transistors. Although the cell provides fast read and write operations, its permanent connection to the power supply causes static power dissipation, particularly during standby conditions.

As low-power VLSI applications continue to grow, reducing the power consumption of memory circuits has become an important design objective. One effective technique for reducing standby power is power gating, in which additional sleep transistors are inserted in the supply path of the circuit [4], [5].

This work presents the design and simulation of a low-power 6T SRAM cell using the power-gating technique in Cadence Virtuoso. The conventional 6T SRAM structure is combined with a PMOS header and an NMOS footer sleep transistor that control the supply and ground connections of the cell. In active mode the sleep transistors conduct and allow normal read, write and hold operations. In sleep mode they are switched OFF to reduce the leakage current flowing through the memory cell.

The circuit is implemented at transistor level and simulated with the Spectre simulator. Transient analysis is used to examine the storage-node voltages, the sleep-control signal and the virtual supply voltage, and the average power is measured with the Cadence ADE Calculator. The measured average power reduces from 942.892 nW to 565.992 nW between the active-mode and sleep-mode simulation runs, and the power dissipated in the sleep transistor reduces from 16.555 nW to 352.004 pW. These results illustrate the potential of power gating to lower power consumption while preserving normal SRAM operation in active mode.

Keywords: 6T SRAM, power gating, sleep transistor, leakage power, virtual supply, Cadence Virtuoso, low-power VLSI.

1.2 Objectives

The main objectives of this work are:

1. **6T SRAM cell design:** to design a conventional 6T SRAM cell using six MOS transistors in Cadence Virtuoso.
2. **Power-gating integration:** to incorporate sleep transistors into the SRAM architecture to control the power supply during active and standby modes.
3. **Active-mode operation:** to verify the read, write and hold operations of the SRAM cell when the sleep transistors are ON.
4. **Sleep-mode operation:** to investigate the reduction in standby power obtained by switching the sleep transistors OFF.
5. **Power measurement:** to measure and compare the average power consumption of the SRAM cell in active and sleep modes.
6. **Cadence simulation:** to perform transient analysis using Cadence Virtuoso and to evaluate the behaviour of the proposed circuit.

1.3 Introduction to SRAM

Static Random Access Memory is a volatile memory that stores binary information using bistable circuits. Each memory cell stores one bit, either logic 0 or logic 1. Unlike DRAM, SRAM does not need periodic refreshing, because the stored value is maintained by the feedback of cross-coupled inverters for as long as the supply voltage is available [1], [2].

A conventional 6T SRAM cell consists of:

- two PMOS transistors forming the pull-up network,
- two NMOS transistors forming the pull-down network, and
- two NMOS access transistors connecting the cell to the bit lines.

The six transistors form a bistable storage circuit with complementary internal nodes Q and QB.

1.3.1 Power Consumption in SRAM

The total power consumption of a CMOS circuit is the sum of its dynamic and static components:

$$P_{\text{total}} = P_{\text{dynamic}} + P_{\text{static}} \tag{1.1}$$

Dynamic power arises from the charging and discharging of circuit capacitances during switching:

$$P_{\text{dynamic}} = \alpha\, C_{\text{L}}\, V_{\text{DD}}^2\, f \tag{1.2}$$

where α is the switching activity factor, C_L is the load capacitance, V_{DD} is the supply voltage and f is the operating frequency. Static power is caused by leakage current that flows through the transistors even when the circuit is not switching:

$$P_{\text{static}} = V_{\text{DD}} \times I_{\text{leak}} \tag{1.3}$$

In an SRAM, the storage transistors remain connected to the supply during standby. Leakage current therefore continues to flow even when no memory operation is performed. Power gating is introduced to remove this unnecessary standby power.

1.3.2 Power-Gating Technique

Power gating is a low-power design technique in which additional MOS transistors, called sleep transistors, are inserted in the power-supply network of a circuit. These transistors control the connection between the SRAM cell and the actual power rails. Two configurations are commonly used:

- **Header switch:** a PMOS transistor connected between the actual V_{DD} and the virtual V_{DD}.
- **Footer switch:** an NMOS transistor connected between the virtual ground and the actual ground.

In active mode the sleep transistors are ON and the circuit operates normally. In sleep mode they are OFF and isolate the SRAM circuit from one or both supply rails, which reduces standby leakage. The proposed design uses sleep-transistor-based power control to study the difference in power consumption between active and sleep modes.

1.4 Organisation of the Report

The report is organised as follows. Chapter 2 describes the system architecture, the working principles and the design workflow. Chapter 3 presents the theoretical background of the 6T SRAM cell and of power gating. Chapter 4 explains the Cadence Virtuoso implementation, including the schematic, test bench and simulation set-up. Chapter 5 presents and discusses the simulation results. Chapter 6 lists the advantages, disadvantages and limitations of the design, and Chapter 7 gives the conclusion and the future scope of the work.

CHAPTER 2

ALGORITHM AND WORKFLOW

2.1 System Architecture

The proposed system consists of a conventional 6T SRAM cell integrated with a power-gating network. The SRAM cell stores one bit of information using two cross-coupled inverters. Two access transistors connect the internal storage nodes to the bit lines during read and write operations. The power-gating network consists of sleep transistors driven by a sleep-control signal, which regulate the supply connection of the cell.

The major blocks of the proposed architecture are:

1. **6T SRAM cell:** stores one bit of binary information.
2. **Access transistors:** control the connection between the internal storage nodes and the bit lines.
3. **Sleep transistors:** control the power-supply and ground connections of the cell.
4. **Sleep-control signal:** determines whether the SRAM operates in active mode or sleep mode.
5. **Power supply:** provides the required operating voltage.
6. **Output nodes:** Q and QB represent the stored data and its complement.

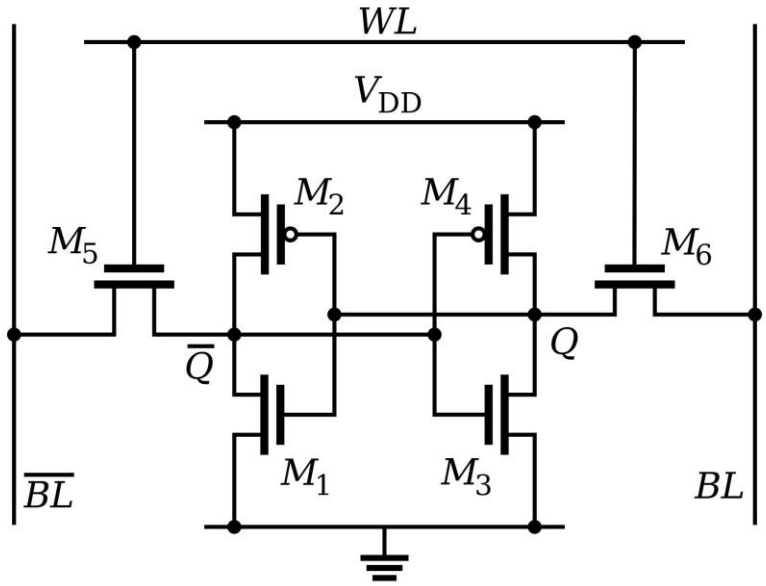


Fig. 2.1: Basic architecture of a conventional 6T SRAM cell.

2.2 Working Principle of 6T SRAM

A conventional 6T SRAM cell uses two cross-coupled inverters to store one bit of data. The two internal nodes are Q, which stores the original data, and QB, which stores the complementary data. The cross-coupled connection provides positive feedback, so the cell retains its information while the supply is available. The cell performs three basic operations:

Write operation. The desired data and its complement are applied to the bit lines BL and BLB. When the word line (WL) is activated, the access transistors turn ON and the bit-line values overwrite the internal storage nodes.

Read operation. The bit lines are first precharged. When WL is activated, the stored value creates a small voltage difference between BL and BLB, which can be detected by a sense amplifier.

Hold operation. WL is LOW, so the access transistors are OFF and the cross-coupled inverters maintain the stored information through feedback.

2.3 Working Principle of Power Gating

Power gating controls the supply connection of the SRAM cell using additional sleep transistors. The operation is divided into two modes.

Active mode

- The sleep-control signal turns the sleep transistors ON.
- The virtual supply approaches the actual supply voltage and the virtual ground approaches the actual ground.
- The SRAM cell receives its operating supply, so read, write and hold operations can be performed.

Sleep mode

- The sleep-control signal turns the sleep transistors OFF.
- The connection between the SRAM cell and the supply network is interrupted.
- The leakage current is reduced, and therefore the power consumed in the standby condition decreases.

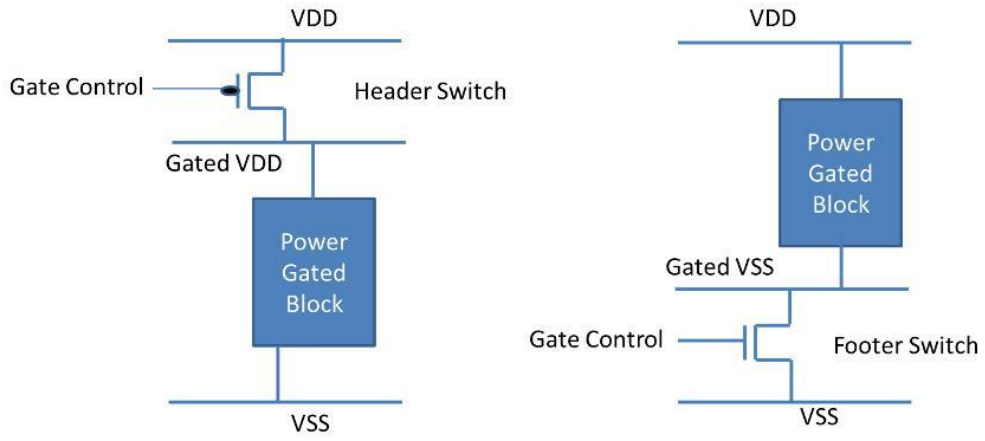


Fig. 2.2: Power-gating architecture using header and footer sleep transistors.

2.4 Input and Control Conditions

Table 2.1 summarises the expected operating conditions of the power-gated SRAM in the two modes.

Table 2.1: Operating conditions in active and sleep modes.

Parameter	Active mode	Sleep mode
Sleep signal S	HIGH (1.8 V)	LOW (0 V)
Header PMOS	ON	OFF
Footer NMOS	ON	OFF
Virtual V_{DD}	Approximately V_{DD}	Isolated from V_{DD}
Virtual GND	Approximately 0 V	Isolated from GND
SRAM operation	Normal operation	Standby
Leakage current	Present	Reduced

In the implemented circuit the footer NMOS transistor is driven by the sleep signal S and the header PMOS transistor is driven by its complement SB. A HIGH level on S therefore selects active mode and a LOW level selects sleep mode.

2.5 Design Workflow

The implementation follows the sequence of steps shown in Fig. 2.3. The 6T cell is designed first, the sleep transistors and control signals are then added, and the circuit is verified by transient simulation. Finally, the average power is measured and the percentage power reduction is calculated.

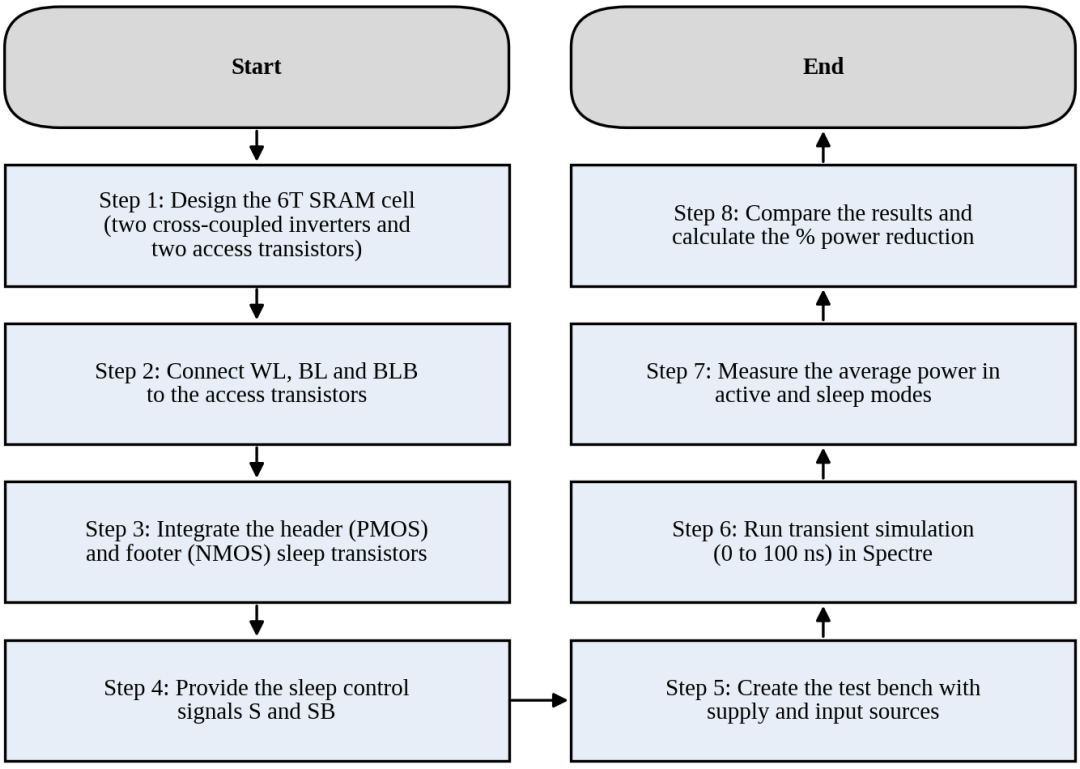


Fig. 2.3: Design and simulation workflow.

CHAPTER 3

THEORETICAL BACKGROUND AND ANALYSIS

3.1 Introduction

Low-power memory design is an important aspect of modern VLSI systems. SRAM cells consume power during read and write operations and also during standby, because leakage currents flow through the MOS transistors even when the memory is idle. Power gating reduces this standby power by controlling the supply connection with sleep transistors. This chapter explains the structure and operation of the 6T SRAM cell, the power-consumption mechanisms, the operation of sleep transistors, and the equations used to evaluate the power reduction.

3.2 Structure of a Conventional 6T SRAM Cell

A conventional 6T SRAM cell consists of six transistors, labelled M1 to M6 in Fig. 2.1 and described in Table 3.1. Four transistors form two cross-coupled CMOS inverters and the remaining two connect the storage nodes to the bit lines BL and BLB. The feedback between the inverters maintains complementary logic values at Q and QB.

Table 3.1: Transistors of the 6T SRAM cell and their functions.

Transistor	Type	Function
M1	NMOS	Pull-down transistor of the first inverter
M2	PMOS	Pull-up transistor of the first inverter
M3	NMOS	Pull-down transistor of the second inverter
M4	PMOS	Pull-up transistor of the second inverter
M5	NMOS	Access transistor connecting BLB to the storage node
M6	NMOS	Access transistor connecting BL to the storage node

3.3 SRAM Cell Operation

3.3.1 Write Operation

To write logic 1, the bit lines are driven to BL = 1 and BLB = 0 and the word line is activated, so that the internal storage nodes change accordingly. To write logic 0, BL = 0 and BLB = 1 are applied. A successful write requires the access transistor to overpower the pull-up transistor of the node that must be pulled LOW. This is expressed by the pull-up ratio:

$$PR = \frac{(W/L)_{\text{load}}}{(W/L)_{\text{access}}} \tag{3.1}$$

3.3.2 Read Operation

The bit lines are initially precharged to a HIGH voltage. When WL becomes HIGH, the access transistors connect the storage nodes to the bit lines. Depending on the stored value, one bit line discharges slightly faster than the other, and the resulting voltage difference represents the stored data. To avoid disturbing the stored value during a read, the pull-down transistor must be stronger than the access transistor. This is expressed by the cell ratio:

$$CR = \frac{(W/L)_{\text{driver}}}{(W/L)_{\text{access}}} \tag{3.2}$$

3.3.3 Hold Operation

When WL is LOW, both access transistors are OFF and the cross-coupled inverters retain the previously stored data. This continues as long as the cell receives a sufficient supply voltage.

3.4 Power Consumption in SRAM

The power consumption of an SRAM is classified into dynamic and static components, as given by Equations (1.1) to (1.3). Dynamic power is consumed when internal nodes, bit lines and word lines switch between logic states. Static power is caused by leakage currents in the MOS transistors, which arise from three main mechanisms [6]:

1. **Subthreshold leakage:** current that flows between drain and source when the gate-to-source voltage is below the threshold voltage.
2. **Gate leakage:** tunnelling current through the thin gate oxide.
3. **Reverse-biased junction leakage:** current through the reverse-biased drain-to-body and source-to-body junctions.

In scaled technologies, subthreshold leakage is usually the dominant component. It is given approximately by

$$I_{\text{sub}} = I_0 \, e^{\frac{V_{\text{GS}} - V_{\text{th}}}{n V_{\text{T}}}} \left(1 - e^{-\frac{V_{\text{DS}}}{V_{\text{T}}}}\right) \tag{3.3}$$

where I_0 is a process-dependent constant, n is the subthreshold slope factor and V_T is the thermal voltage. Because this current depends exponentially on the gate voltage, interrupting the supply path with a high-threshold sleep transistor is an effective way to reduce it. Power gating therefore targets mainly the standby leakage component.

3.5 Power-Gating Technique

Power gating introduces a high-threshold MOS transistor, called a sleep transistor, between the power supply and the circuit or between the circuit and ground. This approach is closely related to the multi-threshold CMOS (MTCMOS) technique [4] and to the gated-V_{DD} technique proposed for cache memories [5]. There are two major configurations (Fig. 2.2).

3.5.1 Header Switch

A PMOS sleep transistor is placed between the actual V_{DD} and the virtual V_{DD}. When it is ON, the circuit is connected to V_{DD}; when it is OFF, the circuit is disconnected from V_{DD}.

3.5.2 Footer Switch

An NMOS sleep transistor is placed between the virtual ground and the actual ground. When it is ON, the circuit is connected to ground; when it is OFF, the circuit is disconnected from ground. When the power-gating transistors are OFF, the internal supply nodes are isolated from the corresponding external rails.

3.6 Working of the Proposed Power-Gated SRAM

The proposed SRAM operates in two modes, compared in Table 3.2.

Table 3.2: Comparison of active and sleep modes.

Operating condition	Active mode	Sleep mode
Sleep transistors	ON	OFF
Power connection	Enabled	Interrupted
SRAM supply	Available	Reduced / isolated
Leakage current	Higher	Lower
Power consumption	Higher	Lower

During active mode, the sleep transistors provide a low-resistance path for the supply current, so the SRAM cell can perform its normal functions. During sleep mode, the sleep transistors

are switched OFF, which increases the isolation between the SRAM cell and the external supply rails and reduces the leakage current.

One important consideration is that a conventional 6T SRAM cell may lose its stored data if its supply is completely disconnected. The data-retention capability of the design must therefore be evaluated separately before it is claimed that the stored information is preserved in sleep mode.

3.7 Theoretical Analysis of Power Reduction

The average power consumption of the circuit over a simulation period T is

$$P_{\text{avg}} = \frac{1}{T}\int_0^T V_{\text{DD}}(t)\, I_{\text{DD}}(t)\, dt \tag{3.4}$$

where P_{avg} is the average power, $V_{DD}(t)$ is the supply voltage and $I_{DD}(t)$ is the current drawn from the supply. For a constant supply voltage, this reduces to

$$P_{\text{avg}} = V_{\text{DD}} \times I_{\text{avg}} \tag{3.5}$$

The percentage power reduction obtained by power gating is calculated as

$$\%P_{\text{reduction}} = \frac{P_{\text{active}} - P_{\text{sleep}}}{P_{\text{active}}} \times 100 \tag{3.6}$$

where P_{active} and P_{sleep} are the average power consumptions in active and sleep modes respectively. The effectiveness of power gating is evaluated by comparing these two values over equivalent measurement intervals.

3.8 Virtual Supply Voltage

The virtual supply voltage (V_{DDV}) is the voltage available to the SRAM cell after the header sleep transistor. When the header transistor is ON,

$$V_{\text{DDV}} \approx V_{\text{DD}} \tag{3.7}$$

When the header transistor is OFF, the virtual supply is isolated and may decrease, depending on the stored charge, the leakage paths and the circuit configuration. Similarly, the footer transistor controls the virtual ground connection. The virtual supply voltage is an important parameter for verifying that the power-gating technique operates correctly.

3.9 Design Considerations

The following factors must be considered in the design of a power-gated SRAM:

1. **Sleep-transistor sizing:** a larger transistor lowers the voltage drop but increases area and its own leakage.
2. **Voltage drop:** the ON resistance of the sleep transistor reduces the voltage available to the cell in active mode.
3. **Standby leakage:** the OFF-state leakage of the sleep transistor itself limits the achievable saving.
4. **Wake-up delay:** time is required to restore the virtual supply when moving from sleep to active mode.
5. **Data retention:** the stored value must be preserved, or restored, after sleep.
6. **Control-circuit power:** the sleep-control circuitry also consumes power.
7. **Transient current:** mode transitions cause switching and in-rush currents.

A proper balance between leakage reduction, active-mode performance and area overhead is necessary for an effective power-gated SRAM design.

CHAPTER 4

CADENCE VIRTUOSO IMPLEMENTATION

4.1 Introduction

The proposed low-power 6T SRAM cell using the power-gating technique is implemented in Cadence Virtuoso. The circuit is designed at transistor level using PMOS and NMOS devices in the Virtuoso Schematic Editor. The conventional 6T SRAM cell is integrated with a PMOS header sleep transistor and an NMOS footer sleep transistor that control the supply connections in active and sleep modes. The circuit is simulated with the Spectre simulator through the Analog Design Environment (ADE), and the waveforms are viewed in the Virtuoso Visualization and Analysis tool.

4.2 Implementation of the Conventional 6T SRAM Cell

The basic SRAM cell is implemented using six MOS transistors: two PMOS transistors form the pull-up network, two NMOS transistors form the pull-down network and two NMOS transistors act as access transistors. The four storage transistors are connected as two cross-coupled CMOS inverters, and the internal nodes Q and QB maintain complementary logic values. The access transistors connect the storage nodes to the bit lines BL and BLB and are controlled by the word-line signal WL. The cross-coupled configuration provides the feedback required to hold the stored information. The schematic is shown in Fig. 4.1.

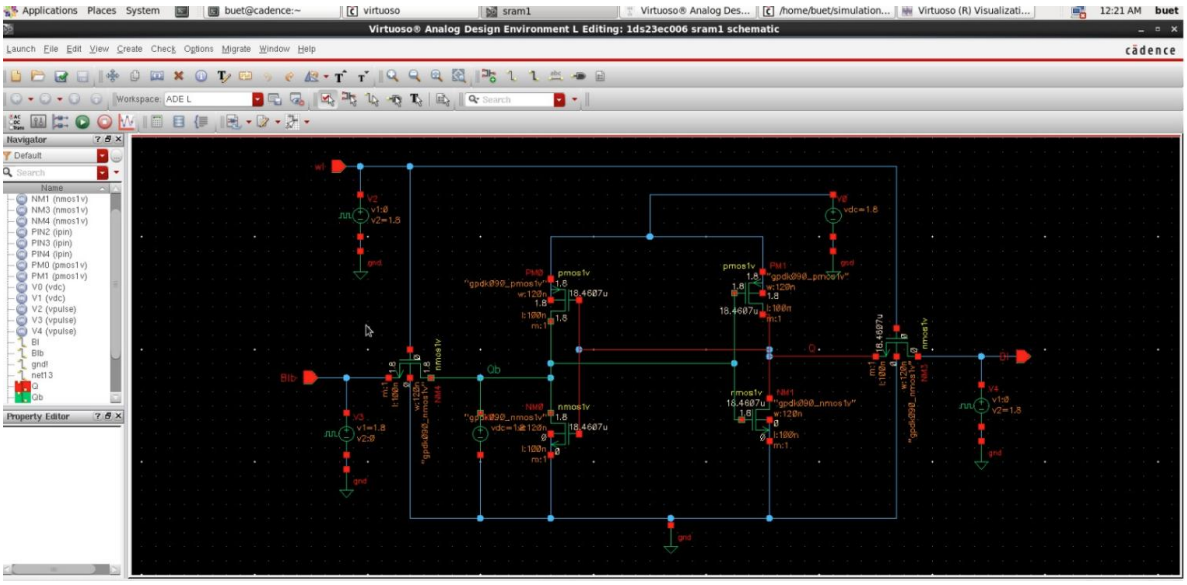


Fig. 4.1: Conventional 6T SRAM cell implementation in Cadence Virtuoso.

4.3 Integration of Power-Gating Transistors

To reduce the standby power consumption, sleep transistors are integrated into the supply network of the conventional SRAM cell, as shown in Fig. 4.2. The following elements are added:

1. **PMOS header sleep transistor (PM5):** controls the connection between the actual supply voltage and the virtual supply voltage of the SRAM cell.
2. **NMOS footer sleep transistor (NM9):** controls the connection between the virtual ground and the actual ground.
3. **Sleep-control circuit:** provides the control signals S and SB that operate the sleep transistors.

The PMOS header transistor is controlled by the complementary sleep signal SB, while the NMOS footer transistor is controlled by the sleep signal S. During active mode, the header and footer transistors are ON and provide the required supply path to the SRAM cell. During sleep mode, both transistors are OFF, which isolates the SRAM circuit from the supply network and reduces the standby leakage.

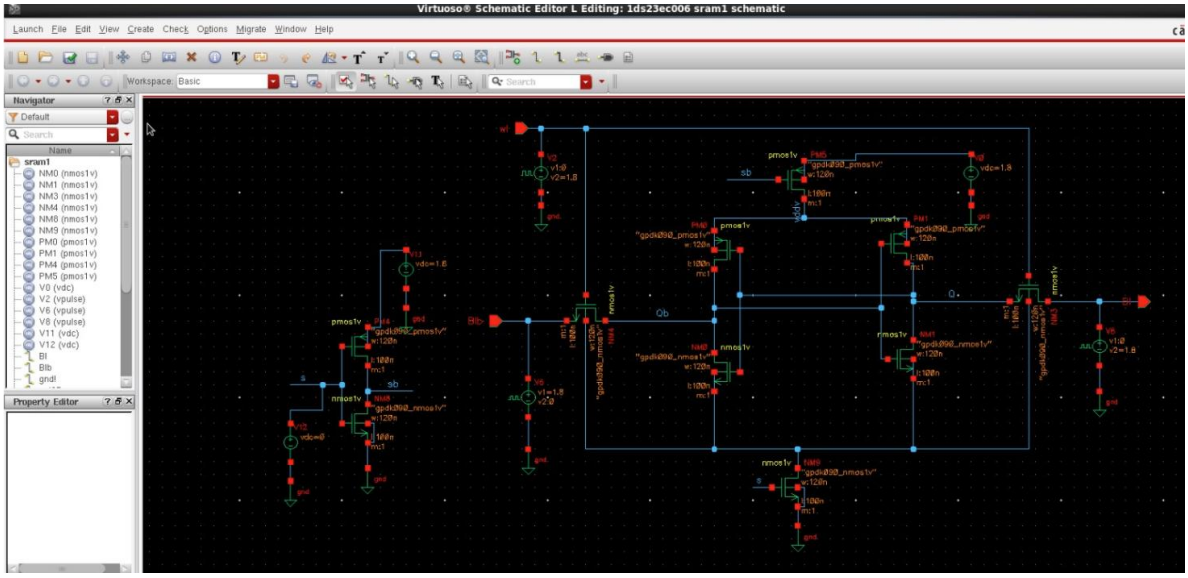


Fig. 4.2: Power-gated 6T SRAM schematic with header and footer sleep transistors.

4.4 Test Bench Implementation

A test bench provides the input signals and operating conditions required by the power-gated SRAM circuit. It consists of voltage sources for the supply voltage, bit lines, word line and sleep-control signals, as shown in Fig. 4.3. The signals used in the simulation are listed in Table 4.1.

Table 4.1: Signals used in the simulation.

Signal	Function
VDD	Main supply voltage of 1.8 V
WL	Word line; controls the access transistors
BL	Bit-line input
BLB	Complementary bit-line input
Sleep (S)	Controls the footer sleep transistor
Sleep bar (SB)	Controls the header sleep transistor
Q	First internal storage node
QB	Complementary storage node
VDDV	Virtual supply voltage

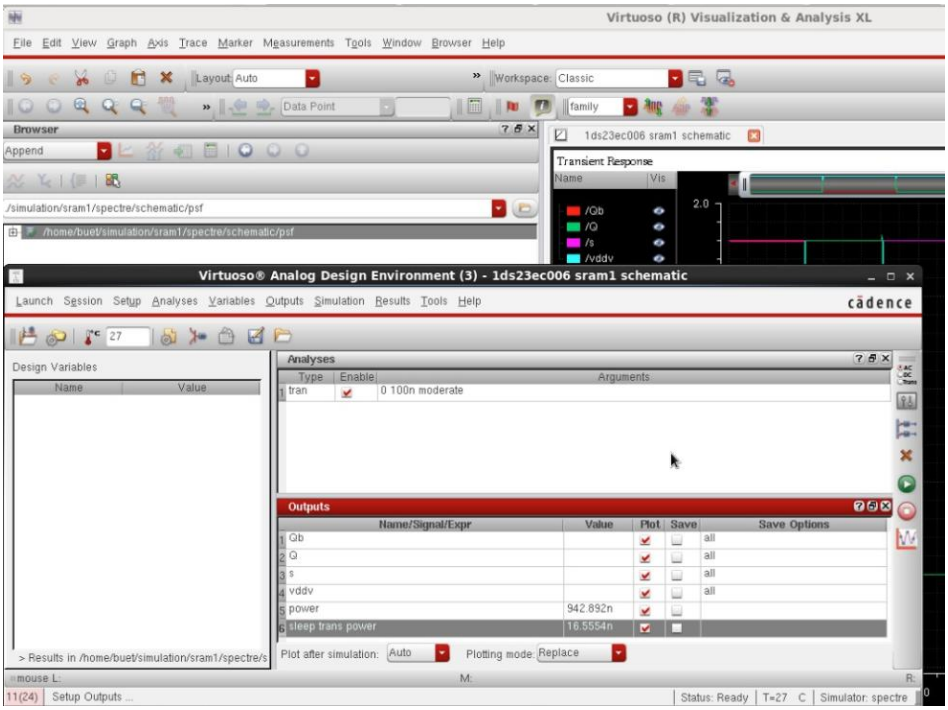


Fig. 4.3: Test bench configuration of the power-gated SRAM in Cadence Virtuoso.

4.5 Transient Analysis Setup

Transient analysis is performed to examine the behaviour of the SRAM circuit with respect to time. The virtual supply voltage and the internal storage-node voltages are monitored during the simulation, so that the response of the cell to changes in the control and input signals can be observed. The settings used are given in Table 4.2.

Table 4.2: Simulation settings.

Parameter	Value
Simulator	Spectre
Analysis	Transient (tran)
Stop time	100 ns
Accuracy setting	Moderate
Temperature	27 °C
Supply voltage	1.8 V
Signals observed	Q, QB, S (sleep), VDDV and instantaneous power

4.6 Average Power Measurement Using the Cadence Calculator

The average power consumption is evaluated using the Calculator available in Cadence Virtuoso. The instantaneous power waveform is obtained from the supply voltage and current measurements, and the average function is applied over the required simulation interval:

$$P(t) = V_{\mathrm{DD}}(t) \times I_{\mathrm{DD}}(t) \tag{4.1}$$

$$P_{\mathrm{avg}} = \mathrm{average}\big(P(t)\big) \tag{4.2}$$

Two output expressions are defined in the ADE Outputs window: *power*, which gives the average power drawn by the circuit, and *sleep trans power*, which gives the power dissipated in the sleep transistor. Their values are displayed in the ADE Outputs table after each simulation run and are reported in Chapter 5.

CHAPTER 5

SIMULATION RESULTS AND DISCUSSION

5.1 Introduction

The proposed power-gated 6T SRAM cell is simulated in Cadence Virtuoso to investigate its electrical behaviour and power consumption. The results include the transient response of the storage nodes, the behaviour of the virtual supply voltage and the average power measurements. The main objective is to verify the operation of the SRAM cell and to evaluate the effect of the sleep transistors on the power consumption of the circuit.

5.2 Transient Response Analysis

Fig. 5.1 shows the transient response over 100 ns. The signals displayed are /Qb (complementary storage node), /pwr (instantaneous power), /Q (storage node), /s (sleep-control signal) and /vddv (virtual supply voltage).



Fig. 5.1: Transient response of Qb, instantaneous power, Q, sleep signal and virtual supply voltage.

In this run the sleep signal /s remains at 1.8 V throughout the simulation, so the sleep transistors stay ON and the circuit operates in active mode. Before 10 ns and after 20 ns, /Qb is at approximately 1.8 V, /Q is close to 0 V and /vddv is approximately 1.8 V, which confirms that the cell holds a stable state with the virtual supply close to the nominal supply. Between about 10 ns and 20 ns, a transient event occurs: /Qb falls to approximately 1.66 V, /Q rises to approximately 0.4 V, /vddv dips to approximately 1.74 V and the instantaneous power

increases. After the event, all the signals return to their original values. The dip of about 60 mV in /vddv is consistent with the voltage drop across the ON resistance of the header transistor while the cell draws a larger current.

Fig. 5.2 shows a second transient simulation in which the word line (WL), the bit lines (BL and BLB) and the storage nodes (Q and QB) are plotted together. The storage nodes respond to the word-line pulse applied through the access transistors, which demonstrates the write path of the cell.

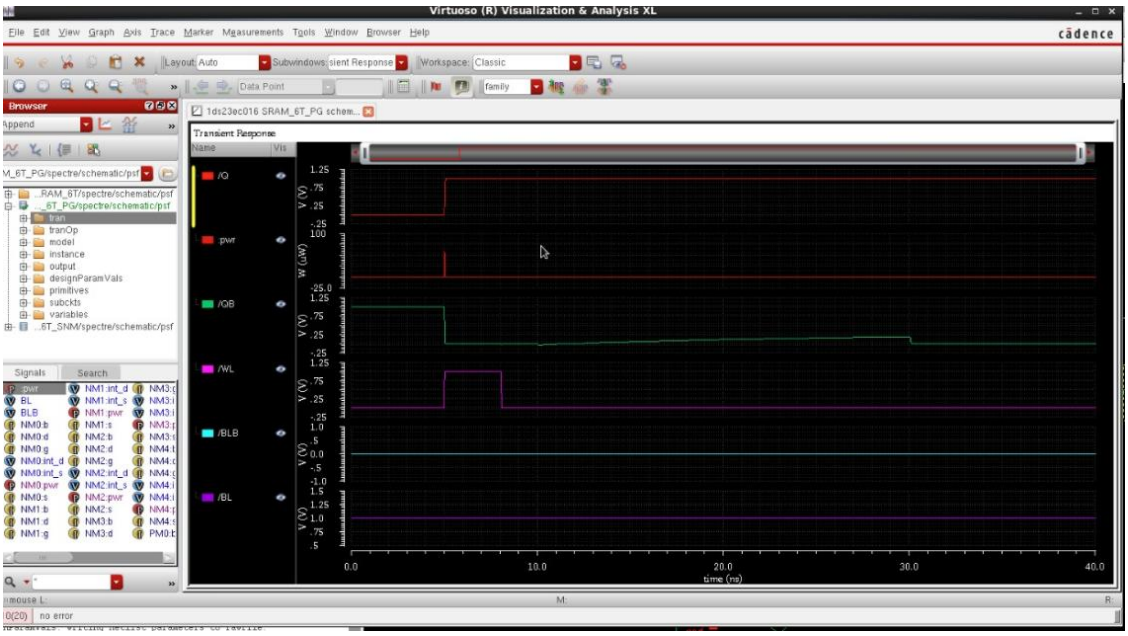


Fig. 5.2: Transient response of the 6T SRAM cell showing Q, QB, WL, BL and BLB.

5.3 Active-Mode Power Measurement

During active mode, the sleep transistors remain ON and the SRAM cell receives its operating supply voltage. The average power is evaluated with the Cadence Calculator over the 0 to 100 ns transient window. The ADE Outputs window for this run (Fig. 5.3) reports an average power of 942.892 nW and a sleep-transistor power of 16.555 nW.

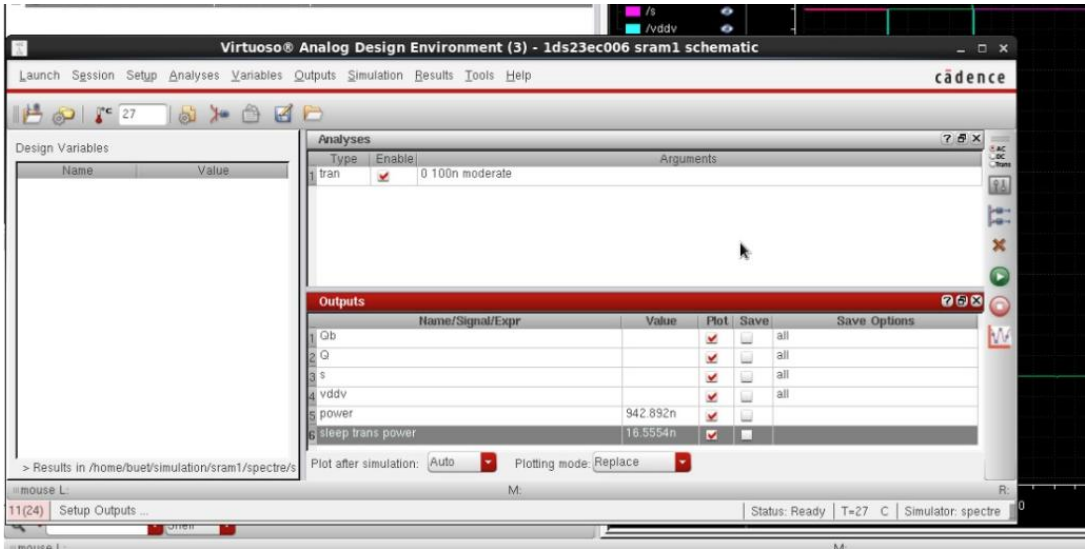


Fig. 5.3: Power measurement in Cadence Virtuoso: active-mode simulation run.

5.4 Sleep-Mode Power Measurement

During sleep mode, the sleep transistors are switched OFF, which reduces the supply current flowing through the SRAM circuit. The ADE Outputs window for this run (Fig. 5.4) reports an average power of 565.992 nW and a sleep-transistor power of 352.004 pW.

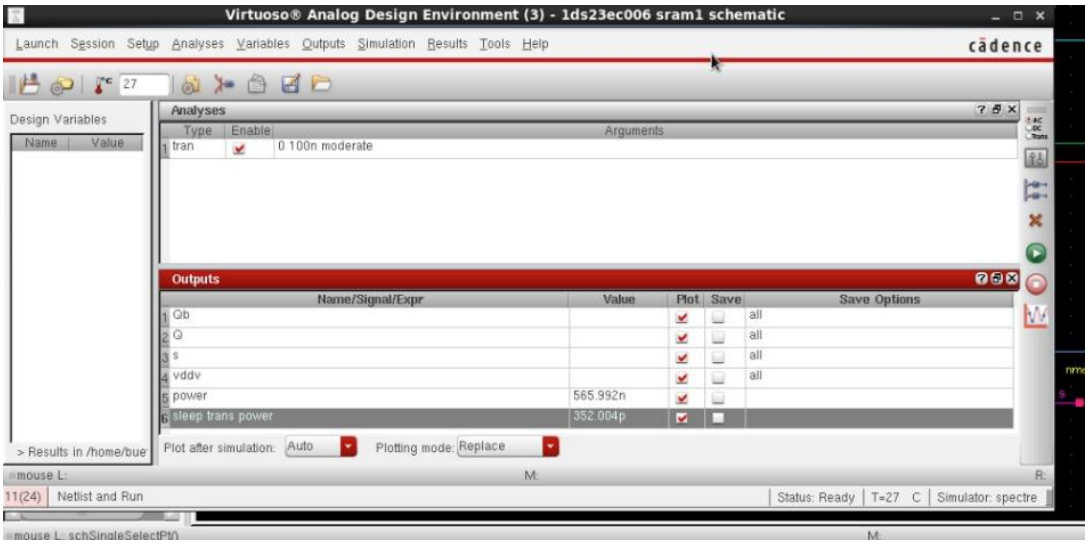


Fig. 5.4: Power measurement in Cadence Virtuoso: sleep-mode simulation run.

5.5 Comparison of Simulation Results

Table 5.1 compares the two measurements, taken at V_{DD} = 1.8 V and 27 °C over the 0 to 100 ns window, and gives the percentage reduction calculated with Equation (3.6).

Table 5.1: Comparison of active-mode and sleep-mode power.

Parameter	Active mode	Sleep mode	Reduction (%)
Average power	942.892 nW	565.992 nW	39.97
Sleep-transistor power	16.555 nW	352.004 pW	97.87

The average power of the circuit decreases by about 40 % when the circuit is moved from active mode to sleep mode. The power dissipated in the sleep transistor itself falls by nearly 98 %, from 16.555 nW to 352.004 pW, which shows that the OFF-state sleep transistor draws very little current.

5.6 Overall Results Discussion

The simulations confirm that the power-gated 6T SRAM operates correctly in active mode: the sleep transistors provide a low-resistance supply path, the storage nodes remain complementary and the virtual supply voltage stays close to 1.8 V, with only a small dip during the transient event. The Calculator measurements show a clear reduction in the average power and in the sleep-transistor power when the sleep transistors are switched OFF.

The average power in sleep mode (565.992 nW) is still large compared with the sleep-transistor power (352.004 pW). The remaining power may be drawn through paths that are not gated, such as the stimulus sources and the control circuitry, and through the remaining leakage of the cell. A more complete evaluation of the standby saving would require measuring the power over an interval in which the circuit is in steady-state sleep, together with the waveform of the virtual supply voltage collapsing during sleep and a check that the stored data are retained or restored on wake-up. These points are listed as future work in Chapter 7.

5.7 Result Summary

The main simulation observations are:

1. The transient simulation shows the storage-node voltages, the sleep-control signal and the virtual supply voltage of the power-gated 6T SRAM.
2. In active mode the virtual supply voltage remains close to the nominal 1.8 V, with a dip of about 60 mV during the transient event.
3. The ADE Calculator gives an average power of 942.892 nW in active mode and 565.992 nW in sleep mode, a reduction of about 39.97 %.

4. The sleep-transistor power reduces from 16.555 nW to 352.004 pW, a reduction of about 97.87 %.
5. Verification of data retention and of the sleep-mode virtual-supply waveform is required to confirm the full standby saving.

CHAPTER 6

ADVANTAGES AND LIMITATIONS

6.1 Advantages

The power-gating technique offers the following advantages for a 6T SRAM cell:

1. **Reduced standby power consumption:** power gating reduces the current drawn in sleep mode by disconnecting the SRAM cell from the supply network.
2. **Improved energy efficiency:** minimising unnecessary power consumption during idle periods improves the energy efficiency of the memory circuit.
3. **Active and sleep mode control:** the sleep-control signal allows the SRAM circuit to switch between active and standby conditions.
4. **Compatibility with the conventional SRAM:** the technique can be integrated with the 6T architecture without changing its basic data-storage principle.
5. **Reduced leakage current:** switching OFF the sleep transistors reduces the unwanted current flow through the memory circuit during standby.
6. **Suitability for low-power applications:** the technique is useful in portable devices, embedded systems and battery-operated applications.

6.2 Disadvantages

The power-gated SRAM also has certain disadvantages:

1. **Additional transistor count:** the sleep transistors increase the total number of transistors in the circuit.
2. **Increased area:** the sleep transistors and control circuitry increase the silicon area.
3. **Wake-up delay:** when switching from sleep to active mode, the virtual supply voltage requires time to reach its normal level.
4. **Voltage drop:** the ON resistance of the sleep transistors may cause a voltage drop in the supply path during active operation.
5. **Control-circuit complexity:** additional sleep-control signals are required to manage the operating modes.
6. **Transient current:** switching between modes may produce temporary current spikes due to the charging and discharging of internal capacitances.

6.3 Limitations

The present work has the following limitations:

1. **Data retention:** if the supply is completely disconnected in sleep mode, the stored information may be lost. Additional retention circuitry may be required if the data must be preserved.
2. **Sleep-transistor sizing:** improper sizing can affect the voltage supplied to the cell and its active-mode performance. A sizing sweep has not been carried out.
3. **Virtual supply voltage:** the virtual supply may not fall as expected if there are unintended connections or leakage paths in the circuit.
4. **Wake-up time:** the time needed to restore the virtual supply has not been measured.
5. **Technology dependence:** the performance of power gating depends on the transistor characteristics and the technology library used.
6. **Simulation scope:** the work is limited to transistor-level simulation of a single cell at one supply voltage and temperature. A full memory array, layout verification, process-corner analysis and physical design are not covered.
7. **Power-measurement accuracy:** an accurate comparison requires measuring the average power over equivalent steady-state intervals and accounting for all supply paths.

CHAPTER 7
CONCLUSION AND FUTURE SCOPE

7.1 Conclusion

This work presented the design and simulation of a low-power 6T SRAM cell using the power-gating technique in Cadence Virtuoso. The conventional 6T SRAM cell, consisting of two cross-coupled inverters and two access transistors, was combined with a PMOS header and an NMOS footer sleep transistor to control the supply and ground connections of the cell.

The circuit operates in two modes. In active mode the sleep transistors are ON and provide the supply connections required for normal SRAM operation. In sleep mode the sleep transistors are OFF and isolate the memory circuit from the supply network, which reduces the standby current. The circuit was simulated with Spectre, and the ADE Calculator was used to measure the average power.

The measurements show that the average power decreases from 942.892 nW to 565.992 nW (about 39.97 %) and the sleep-transistor power decreases from 16.555 nW to 352.004 pW (about 97.87 %) between the active-mode and sleep-mode runs. The results indicate that power gating is an effective technique for reducing the power consumption of SRAM circuits, at the cost of additional transistors, area and control complexity.

7.2 Future Scope

The work can be extended in the following directions:

- analysis of data retention and of the static noise margin of the cell during hold, read and write [7],
- measurement of the wake-up time and of the in-rush current during the sleep-to-active transition,
- a parametric sweep of the sleep-transistor width to find the best trade-off between voltage drop, area and leakage,
- simulation across process corners, supply voltages and temperatures,
- data-retention techniques such as a reduced (non-zero) sleep voltage, and
- layout design with post-layout simulation, and extension to a complete SRAM array.

CHAPTER 8

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